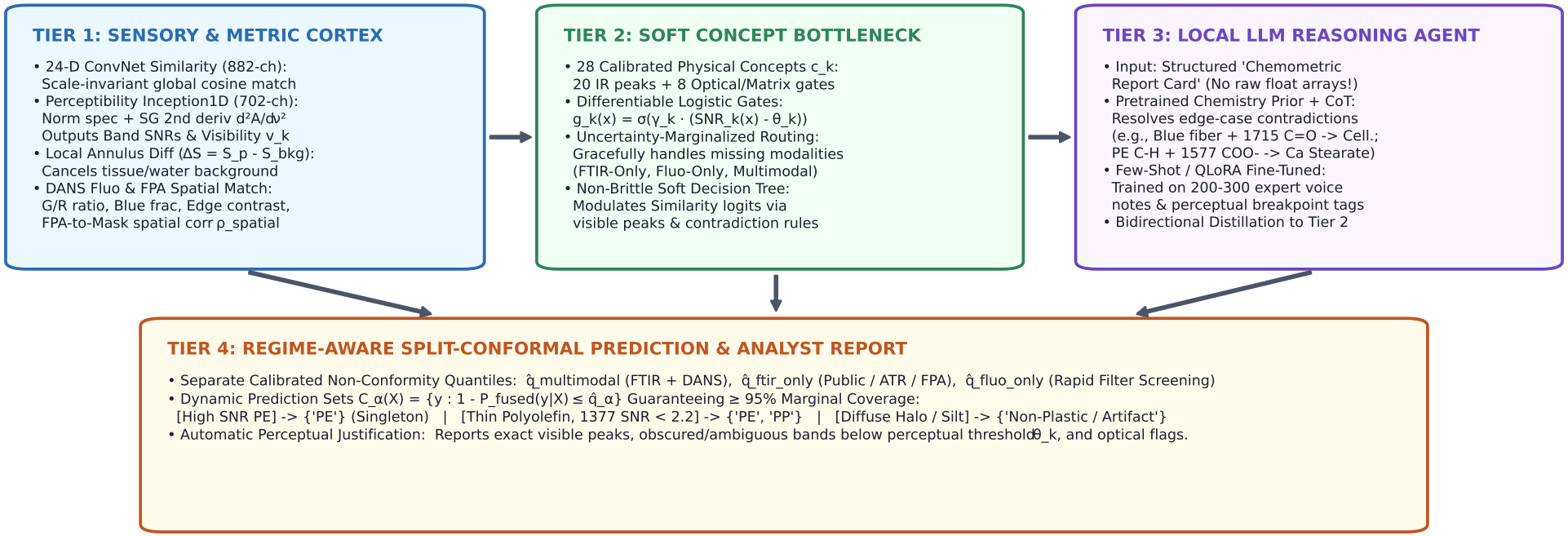
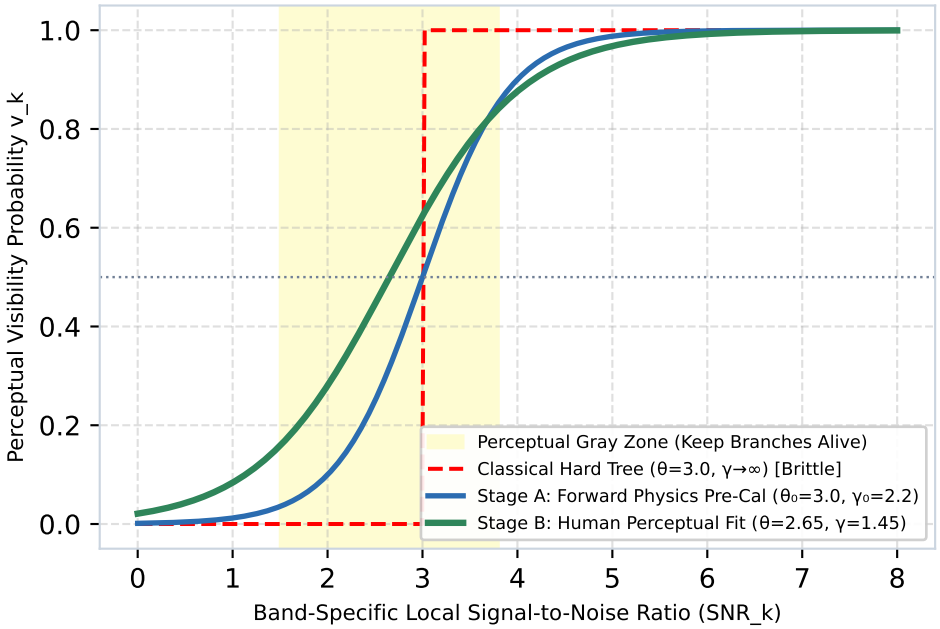


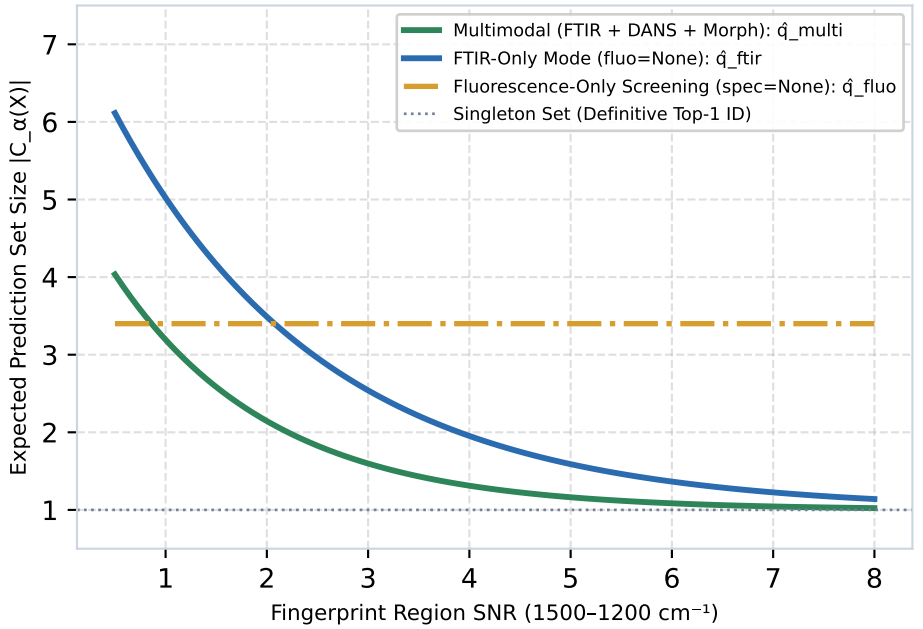
Standalone vs. Hybrid Module Evaluation & Ablation Strategy (Testing Each Component Independently & Jointly)

Module / Configuration	Input Grid & Modalities	Primary Strength	Known Standalone Failure Mode	How Hybrid Architecture Fixes It	Target Benchmark
1. 24-D Similarity ConvNet (Standalone Baseline)	882-ch FTIR (3996–1200) Zero-padded to 896-ch	Fast, scale-invariant global cosine peak-ratio matching	Zero Non-MP awareness; forces silt/tissue/stearate into plastic	Anchored at leaf nodes; gated by Tier 1 Non-MP & Perceptual SNRs	Spiked Filter 1 & Open Specy Pristine
2. Forward Perceptibility Inception1D (Standalone)	702-ch FTIR (4000–1196) [Norm Spec + SG 2nd Deriv]	Quantifies exact band SNR & human visibility $v_k \in [0,1]$	Detects individual peaks but lacks holistic polymer envelope prior	Supplies calibrated visibility weights v_k to Soft Concept Bottleneck	Psychophysical Trials (perceptibility_app)
3. Soft Concept Bottleneck (CBM Tree Standalone)	20 Spectral Concepts + 8 DANS/Morph Concepts	100% explainable rules; smooth logistic gates avoid cliff effects	Cannot invent rules for unseen contaminant combinations	Guided by Similarity anchor and distilled LLM edge-case rules	2,127 Real Annotated Environmental MPs
4. Local LLM Reasoner (Qwen-2.5 / Llama-3.1 8B)	Structured Chemometric Report Card + Voice CoT	Rich pretrained chemistry prior; resolves complex contradictions	Slower inference (~0.3s/particle); raw token logits need calibration	Consumes Tier 1 concepts; wrapped in Tier 4 Split-Conformal calibration	Curated Cohort 24 & 500 Edge-Case Queue
5. Full Hybrid CBM + Sim + Conformal Wrapper	Supports Multimodal, FTIR-Only & Fluo-Only	Combines global shape, local peak SNR, optical stain & chemistry	Requires regime-specific quantile $\hat{q}_\alpha$ for partial modalities	Maintains 3 calibrated quantiles ($\hat{q}$_multi, $\hat{q}$_ftir, $\hat{q}$_fluo) for ≥95% cov.	All Internal Cohorts + Public MP Libraries

Perceptual Gate Calibration: g_k(x) = σ(γ_k · (SNR_k - θ_k))



Dynamic Conformal Set Size vs. Fingerprint SNR (95% Coverage)



1. TWO-STAGE TEMPERATURE (γ_k) & THRESHOLD (θ_k) CALIBRATION

- Soft Gate Definition:
  $g_k(x) = 1 / [1 + \exp(-\gamma_k \cdot (S_k(x) - \theta_k))]$
  where $S_k(x)$ is the Inception1D / SG 2nd-deriv prominence SNR of diagnostic band k.

- Stage A (Zero-Label Physics Pre-Cal via Forward Model):
  Synthesize 10,000 distorted spectra across τ (thickness), ρ_0 (Mie), BSR (bkgd mismatch), and 1/f noise. Fit initial $(\theta_k^{(0)}, \gamma_k^{(0)})$ against true underlying peak presence (Rose detection criterion $\theta_k^{(0)} \approx 2.8–3.2\ \sigma$).

- Stage B (Human Perceptual Fit on 20–30 Trials/Feature):
  Given human tags $y_tag \in \{0 \text{ (Absent)}, 0.5, 1 \text{ (Visible)}\}$, optimize only the 2 scalars (θ_k, γ_k) via Platt scaling:
  $\min_{(\theta_k, \gamma_k)} -\sum [y_tag \log g_k + (1-y_tag) \log(1-g_k)] + \lambda ((\theta_k - \theta_k^{(0)})^2 + (\gamma_k - \gamma_k^{(0)})^2)$
  Because each gate has only 2 parameters, 25–30 real trials lock in the exact human boundary without overfitting!

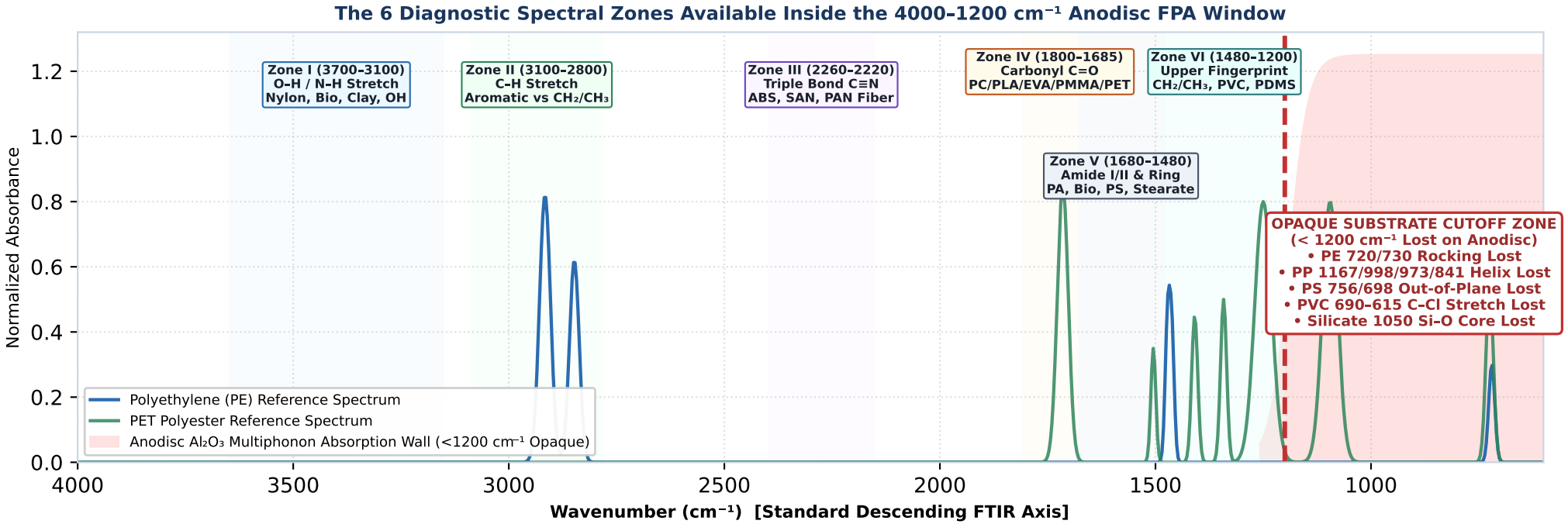
2. UNCERTAINTY-MARGINALIZED ROUTING & MISSING MODALITIES

- Why Classical Decision Trees Break:
  Hard splits (`if ratio_G_R > 1.2`) fail if fluorescence is absent (FTIR-only public datasets) or if water/biofilm obscures the 1640 cm^{-1} window.

- Availability & Visibility Weighting $v_k \in [0, 1]$:
  Each concept gate is marginalized by observability v_k:
  $\tilde{g}_k(x) = v_k \cdot g_k(x) + (1 - v_k) \cdot \pi_k$
  where π_k is the neutral prior (0.5). When fluo=None ($v_fluo = 0$) or when local saturation clips a region ($v_k \to 0$), $\tilde{g}_k$ smoothly relaxes to π_k.

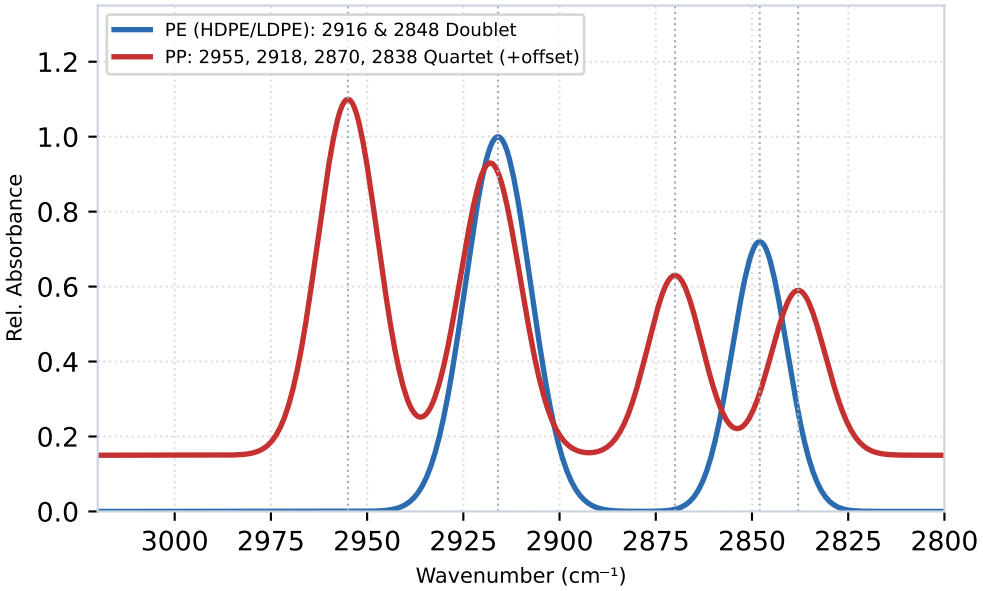
- Hybrid Logit Scoring Anchored by 24-D Cosine Similarity:
  $\text{Logit}(y|x) = \alpha_y \cdot \text{CosSim}(z_{sim}(x), \mu_y)$ [Shape Anchor]
  $+ \sum_k W_{(y,k)} \cdot v_k \cdot (g_k - 0.5)$ [Visible Peaks]
  $- \sum_j B_{(y,j)} \cdot v_j \cdot r_j(x)$ [Contradictions]
  If fingerprint peaks are buried in noise ($v_k \to 0$), PE and PP receive nearly equal logits, expanding C_α to {'PE','PP'}!

Phase / Step	Target Script / Asset Path	Technical Action & Verification Criteria	Sample Target	Status / Gate
1.1 Spectral Grid Standardization	orig_wavenumbers.npy (882-ch) DEFAULT_WN_GRID (702-ch) Transformer Grid (840-ch)	Verify all FPA Anodisc spectra are interpolated strictly to 4000–1200 cm^{-1}; apply (7,7) zero-pad to 896-ch for 24-D ConvNet and SG 2nd deriv (w=15, p=2) for 702-ch Perceptibility Inception1D.	All HDF5 stores (23 queues)	[✓] Verified in codebase
1.2 Local Annulus Bkgd Differential	fpa_data.py & *_correspondences.h5	Extract both inside-mask particle spectrum S_p and 3–5 px dilated local annulus ring S_annulus. Compute differential ΔS = S_p - S_annulus to flag background tissue Amide (1650/1545) and silt leakage.	2,127 annotated + 23 queues	[ ] Add ΔS channel to extractor
2.1 Fluo & Spatial QC Curation	D:\user_data\smolen\fluo_dataset & extract_fluo_features.py	Compute (1) Edge Contrast Ratio (I_core - I_halo)/I_halo to reject diffuse YOLO halos; (2) Blue Fraction B/(R+G+B) for cellulose fibers; (3) Spatial correlation ρ_spatial between Fluo mask and FPA chemical band map.	1,916 fluo crops (curate ~100 bad)	[ ] Run rapid curation pass
2.2 Dedicated Non-MP Exemplar Pools	non_mp_clustered_spectra.npy & particle_notes.json	Isolate explicit negative training pools: (a) 'Green Dot' 1800–1200 cm^{-1} hump, (b) Blue cellulose fibers, (c) Ca/Zn Stearate (2916/2848 + 1577/1540 cm^{-1}), (d) Biological tissue/keratin (1652/1545/1400), (e) Calcite & Silicate ramps.	50–100 exemplars per negative class	[ ] Tag via Web Annotator (`T`)
3.1 Forward Physics SNR Titration	parametric_distortion.py & perceptibility_app/engine.py	Generate synthetic breakpoint curves across all 8 degradation modes (Mie ρ_0, optical saturation τ, under/over-subtraction, biofilm, pink noise, cutoff ramp) for all 28 diagnostic concepts to initialize Stage A ($\theta_k^{(0)}$, $\gamma_k^{(0)}$).	10,000 synthetic titration spectra	[✓] 7 features done; expand to 28
3.2 Psychophysical Breakpoint Trials	perceptibility_app/server.py (port 8055) -> trials.jsonl	Annotate human visibility breakpoints (Visible / Ambiguous / Invisible) on degraded polymer & subtype pairs (HDPE vs LDPE 1377, PA6 vs PA6,6, PVC 1427, PET vs Cellulose) to fit Stage B Platt parameters (θ_k, γ_k).	25–35 trials per feature (~700 tot)	[ ] In progress in Perceptibility App
4.1 Expert Voice & Auto-Tag Cohort	web_annotator/server.py & particle_notes.json	Record 10–20s speech-to-text reasoning notes + hotkey diagnostic tags (`1`–`6`, `A` auto-tag) across curated_cohort_24, new200, and real_samples_edge_cases_500, focusing on PMMA, PC, PVC, Nylon, and weathered polyolefins.	200–300 voice + tag annotations	[ ] Speech-to-text agent integration
4.2 Local LLM CoT Alignment & Distill	Chemometric Report Card -> Qwen-2.5-7B / Llama-3.1-8B	Format Tier 1 outputs into structured JSON Report Cards. Benchmark 0-shot/few-shot rule prompting vs. QLoRA fine-tuning on voice notes. Distill learned contradiction weights B_(y,j) back into fast Tier 2 Soft Concept Tree.	200 train / 100 held-out test	[ ] Build Report Card formatter
5.1 3-Regime Split-Conformal Calibration	run_model_fusion_exploration.py & live_classifier.py	Perform 5-fold cross-conformal calibration (α = 0.05) separately for: (1) Multimodal (FTIR + DANS), (2) FTIR-Only (masked fluo), and (3) Fluo-Only (masked FTIR). Validate OOD FTIR-only generalization on Open Specy / SLiMP.	≥ 200 cal samples per regime	[ ] Calibrate 3 regime quantiles

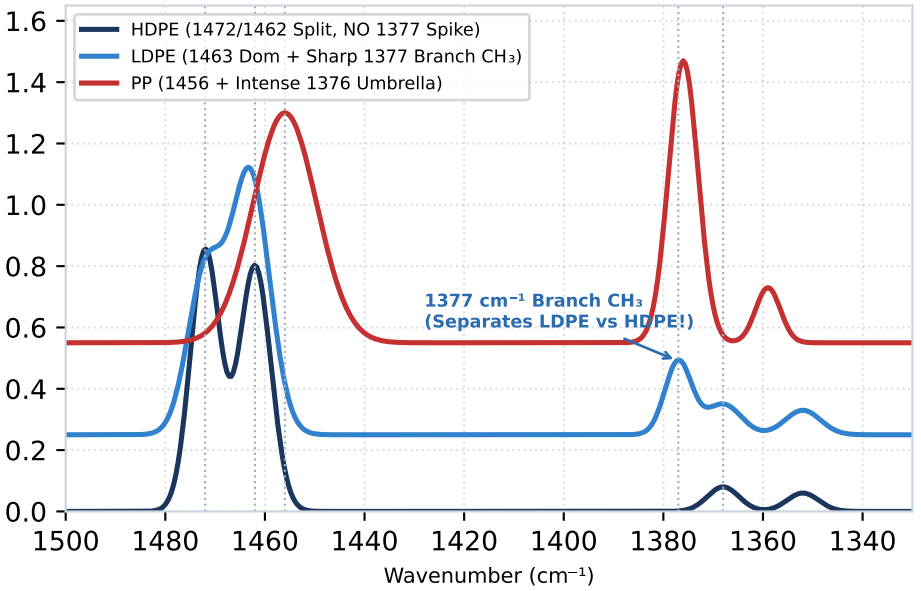


Polymer / Class	Classical Band Lost (<1200 cm^{-1})	Mandatory Surrogate Markers in 4000–1200 cm^{-1}	Quantitative Surrogate Decision Rule
Polyethylene (HDPE vs LDPE)	730 / 720 cm^{-1} orthorhombic CH_2 rocking doublet	• 1472 / 1462 cm^{-1} CH_2 scissoring crystalline doublet • 1377 cm^{-1} branch CH_3 umbrella vs 1368/1352 cm^{-1} wagging	HDPE: A_1377/A_1368 ≤ 1.0 & A_1472/A_1462 > 0.90 LDPE: sharp 1377 spike (A_1377/A_1368 > 1.5)
Polypropylene (Isotactic PP)	1167, 998, 973, 841 cm^{-1} isotactic helix progression	• 2955 / 2918 / 2870 / 2838 cm^{-1} C–H stretch quartet • 1376 cm^{-1} sym CH_3 umbrella + 1456 cm^{-1} asym bend	A_1376 / A_1456 > 0.85 AND A_2955 / A_2918 > 1.05 (plus weak 1359/1329/1304/1256 helix quartet)
Polystyrene (PS / ABS / SAN)	756 & 698 cm^{-1} mono-substituted ring out-of-plane C–H bends	• 3082, 3060, 3026 cm^{-1} aromatic C–H stretch fingers • 1945, 1871, 1802, 1745 cm^{-1} overtone fingers • 1601, 1583, 1493, 1452 cm^{-1} sharp ring quadrant/semicircle	Sharp 1493 & 1452 cm^{-1} spikes (FWHM < 12 cm^{-1}) + 3060/3026 cm^{-1} aromatic C–H stretches
Polyvinyl Chloride (uPVC & pPVC)	690–615 cm^{-1} C–Cl stretching doublet	• 1427 cm^{-1} Cl-inductive shifted CH_2 scissoring (!) • 1332 & 1254 cm^{-1} CH–Cl wagging/bending doublet • 2972 cm^{-1} high-frequency CH–Cl methine stretch	A_1427 / A_1465 > 2.0 (scissor shifted from 1465 to 1427 cm^{-1}!) + strong 1254 & 1332 cm^{-1} peaks
Silicone / PDMS Elastomer	1090–1020 cm^{-1} Si–O–Si doublet & 800 cm^{-1} Si–C stretch	• 1260 cm^{-1} razor-sharp symmetric Si–CH_3 umbrella spike • 1412 cm^{-1} asym Si–CH_3 bend + 2963/2905 cm^{-1} C–H	Needle-sharp spike at 1260±2 cm^{-1} (FWHM < 8 cm^{-1}) with A_1260 / A_1412 > 3.5 + steep <1220 rise
Cellulose vs. Silicates / Quartz	1160–1030 cm^{-1} pyranose C–O & 1050 cm^{-1} Si–O–Si stretch	• Cellulose: 3340 broad OH + single 2898 C–H + 1428/1370/1335/1315 • Clays/Quartz: 3695/3620 sharp Al–OH, 1995/1870 Si–O overtones, steep monotonic cutoff ramp <1260 cm^{-1}	Cellulose: single C–H at 2898 (no 2916/2848 doublet) Silicate: A_1210/A_1300 > 2.5 + zero C–H stretch

C–H Stretching Region (3020–2800 cm⁻¹): PE Doublet vs. PP Quartet

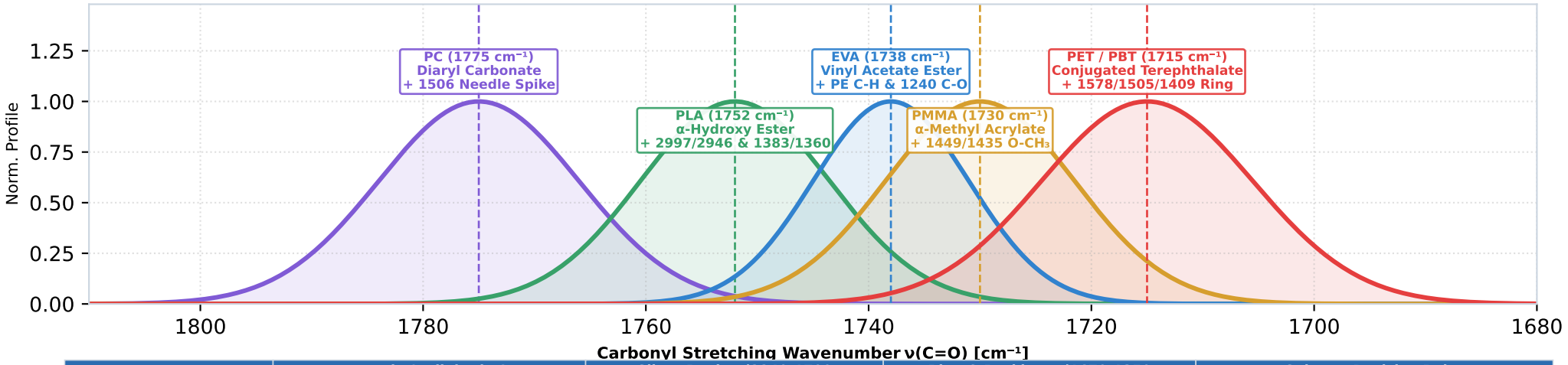


Bending & Wagging Region (1500–1330 cm⁻¹): Subtype Separation



Polymer / Subtype	Primary C–H Stretch (3000–2800 cm⁻¹)	Scissoring & Bending (1480–1440 cm⁻¹)	Umbrella & Wagging (1390–1340 cm⁻¹)	Quantitative Decision Rules & Breakpoint Thresholds
HDPE (High-Density PE, Linear)	2916 (vs, ν_as CH_2) 2848 (vs, ν_s CH_2) [A_2955/A_2916 < 0.10]	1472 & 1462 sharp orthorhombic doublet (A_1472/A_1462 ≈ 0.90–1.20)	Only weak 1368 & 1352 gauche CH_2 wag doublet; NO central 1377 spike!	• R_bend = A_1377 / A_1463 < 0.05 • Branch Ratio A_1377 / A_1368 ≤ 1.0 (trough at 1377) • High crystalline split A_1472 / A_1462 ≥ 0.90
LDPE (Low-Density PE, Branched)	2916 (vs, ν_as CH_2) 2848 (vs, ν_s CH_2) Weak 2955 shoulder	1463 dominant peak with 1472 weaker shoulder (A_1472/A_1463 ≈ 0.55–0.80)	Sharp 1377–1378 cm^{-1} CH_3 umbrella spike protruding above 1368/1352 shoulders	• R_bend = A_1377 / A_1463 ∈ [0.12, 0.25] • Branch Ratio A_1377 / A_1368 > 1.5 (typically 1.8–2.5) • Unweathered weak vinylidene C=C at 1642 cm^{-1}
LLDPE (Linear Low-Density Copolymer)	2916 (vs, ν_as CH_2) 2848 (vs, ν_s CH_2)	1472 & 1462 doublet intermediate splitting (A_1472/A_1462 ≈ 0.75–0.95)	Moderate 1377 cm^{-1} peak slightly above 1368 cm^{-1} wagging shoulder	• R_bend = A_1377 / A_1463 ∈ [0.05, 0.12] • Branch Ratio 1.10 ≤ A_1377 / A_1368 ≤ 1.50 • Requires SNR_fingerprint > 4.5 to separate from LDPE
Isotactic PP (Polypropylene)	4-Peak Quartet: 2955 (vs, ν_as CH_3), 2918 (s), 2870 (m), 2838 (m)	1455–1458 cm^{-1} medium-strong asymmetric band (no 1472 splitting)	1376–1378 cm^{-1} intense, needle-sharp δ_s(CH_3) + 1359, 1329, 1304, 1256	• R_bend = A_1376 / A_1456 > 0.85 (typically 0.95–1.35) • C–H Quartet: A_2955 / A_2918 > 1.05 & ν_s(CH_2) at 2838 • If 2955/2918 saturates, rely on A_1376/A_1456
EPR / EPDM & PP/PE Copolymer	Composite: 2918 & 2850 dominate + strong 2952 & 2870 shoulders	1462–1464 cm^{-1} broadish singlet (or weak 1472 in impact copolymer)	1377 cm^{-1} medium-strong single peak (intermediate between LDPE and PP)	• Intermediate R_bend: 0.30 ≤ A_1377 / A_1463 ≤ 0.75 • Tagged as `PP/PE: Overlapping Scissor + Umbrella` • Conformal Set outputs {'PE', 'PP'} or {'PP/PE'}

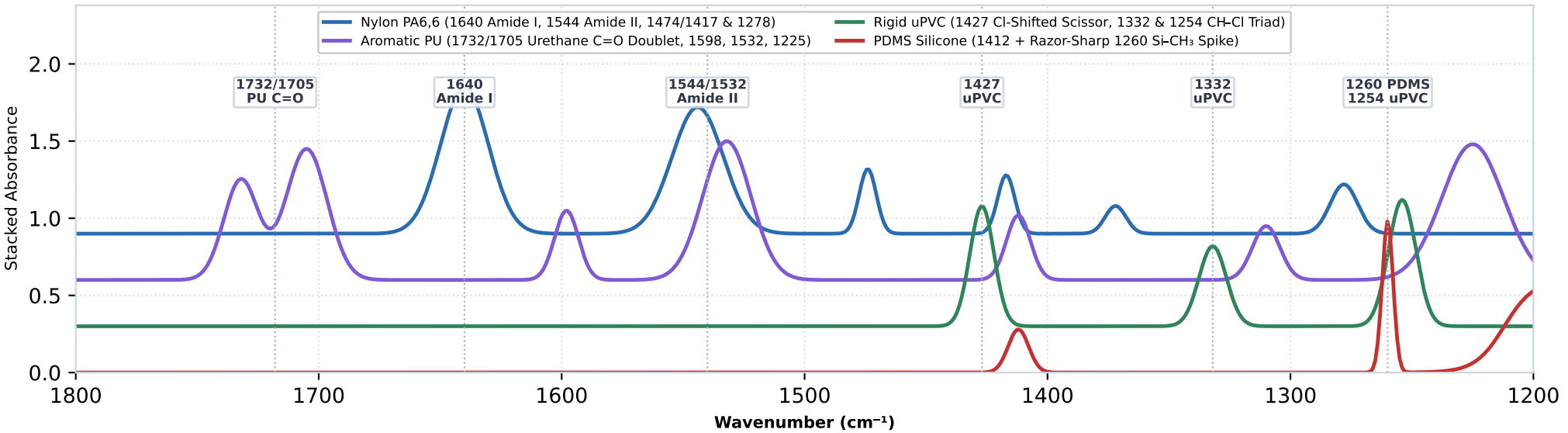
The Carbonyl Thermometer (1800–1690 cm⁻¹): Inductive & Resonance Shifts Separate the 5 Major Ester/Carbonate Classes



Styrenic Subtype	Aromatic & Aliphatic C-H (cm⁻¹)	Silent Region (2240-1700 cm⁻¹)	Ring & Backbone (1670-1350 cm⁻¹)	Subtype Decision Rules (4000-1200 cm⁻¹)
Polystyrene (PS) & HIPS	Aro: 3082, 3060, 3026 (m-s) Ali: 2924 (s), 2850 (m) [A_2924/A_3026 ≈ 0.9–1.3]	NO 2237 peak (R_CN = 0); Weak 'Aromatic Fingers': 1945, 1871, 1802, 1745	1601 & 1583 ring pair; 1493 & 1452 very sharp spikes (A_1493/A_1452 ≈ 1.0)	R_CN = A_2237/A_1493 < 0.05 + sharp 1493/1452 doublet (separates from PET/PC at 1505/1506!).
SAN Copolymer (Styrene-Acrylonitrile)	Aro: 3060, 3026 (m) Ali: 2924, 2852 (m-s) [A_2924/A_3026 ≈ 1.0–1.5]	Sharp 2237–2239 cm⁻¹ C≡N stretch spike! (R_CN ≥ 0.18)	1601, 1583, 1493, 1452 (A_1452/A_1493 ≈ 0.90–1.05; no 1665 butadiene peak)	Sharp 2237 C≡N spike + 1493/1452 styrene ring + low aliphatic ratio (A_2924/A_3026 < 1.55).
ABS Terpolymer (Acrylonitrile-Butadiene-Styrene)	Aro: 3060, 3026 (m) Ali: 2924 (vs), 2850 (s) [A_2924/A_3026 ≈ 1.7–3.2]	Sharp 2237 cm⁻¹ C≡N spike + often weathered 1715 cm⁻¹ butadiene oxidation C=O	1665 (trans-1,4 C=C) & 1640 (vinyl C=C) + 1601, 1493, 1452 (A_1452/A_1493 ≈ 1.15–1.45)	2237 C≡N + 1493/1452 ring + elevated butadiene CH₂ (A_2924/A_3026 > 1.65 & A_1452 > A_1493).

Polymer Class	C=O Center (cm⁻¹)	C-H Stretch Profile (3080–2800 cm⁻¹)	Confirming Bands (1620–1200 cm⁻¹)	Key Separation & Subtype Rules (>1200 cm⁻¹)
PET (Polyethylene terephthalate)	1713–1718 (vs, conj.)	3430 (w, sharp 2×C=O overtone); 3060 (w, aro CH); 2965 & 2908 (w-m, ethylene CH₂)	• 1578, 1505, 1409 (para-ring triad) • 1342 (s, crystalline trans CH₂ wag) • 1245–1265 (vs, massive ester C–O)	PET vs PBT: PET has dominant 1342 cm⁻¹ trans wag (A_1342/A_1453 > 1.3) and weak C–H (A_2965/A_1505 < 1.2). 3430 overtone is sharp (vs broad Cellulose OH!).
PBT (Polybutylene terephthalate)	1711–1716 (vs, conj.)	Richer aliphatic multiplet: 2960, 2925, 2870, 2855 (m-s, butylene -(CH₂)₄- chain)	• 1578, 1505, 1409 (terephthalate triad) • 1460 (m-s, CH₂ scissor) & 1386/1358 • 1250–1270 (vs, ester C–O)	PBT replaces PET's 1342 cm⁻¹ peak with strong 1460 & 1386 cm⁻¹ bands (A_1342/A_1460 < 0.40) and 2.5–4× stronger aliphatic C–H stretches.
PMMA (Polymethyl methacrylate)	1726–1732 (vs, sharp)	Unique O–CH₃ / α-CH₃ triad: 2995 & 2950 (m-s), 2842 (w-m, sharp sym O–CH₃)	• Flat 1650–1500 (ZERO aromatic bands!) • 1485, 1449/1435 (s, O–CH₃ doublet) • 1388 (m, α-CH₃) + 1270 & 1242 (s, C–O)	1730 C=O + 2995/2950/2842 C–H + 1449/1435 doublet with zero aromatic bands. (Priority class for additional real particle annotations!).
EVA (Ethylene-Vinyl Acetate)	1736–1740 (s-vs, narrow, FWHM < 22)	2918 & 2850 (vs, PE-like doublet) + 3460 & 2678 (vw, sharp C=O overtone)	• 1465 (m-s, ethylene CH₂ scissor) • 1372 (m-s, sharp acetate CH₃ umbrella) • 1240 (s, broad acetate C–O–C stretch)	VA% ratio R_EVA = A_1738/A_1465 (0.3–4.5). Separates from weathered PE via narrow 1738 C=O + proportional 1240 C–O band (A_1240/A_1738 > 0.45).
PC (Bisphenol-A Polycarbonate)	1770–1777 (vs, highest polymer C=O)	3060, 3040 (w, aro CH); 2968 & 2873 (m, isopropylidene gem-dimethyl CH₃)	• 1506 (vs, needle-sharp para-ring spike!) • 1600 (m) & 1409 (m) • 1386 & 1364 (gem-CH₃ doublet) + 1230	Unmistakable high-ν 1775 cm⁻¹ carbonate C=O paired with needle-sharp 1506 cm⁻¹ ring spike (A_1506/A_1775 > 0.5) and 1386/1364 cm⁻¹ Bisphenol-A doublet.
PLA (Polylactic Acid Bioplastic)	1748–1756 (vs, aliphatic α-ester)	2997 & 2946 (m, sharp CH₃); 2880 (w, methine C–H) [NO strong 2918/2850 CH₂!]	• Flat 1650–1500 (zero aromatic/amide) • 1453 (m-s, δ_as CH₃) • 1383 & 1360 (m-s doublet) + 1268/1212	1752 cm⁻¹ C=O + 2997/2946 C–H + 1453 & 1383/1360 doublet with absence of 2918/2850 CH₂ doublet.

Upper-Fingerprint Signatures (1800–1200 cm⁻¹) Separating Polyamides, Polyurethanes, PVC & PDMS



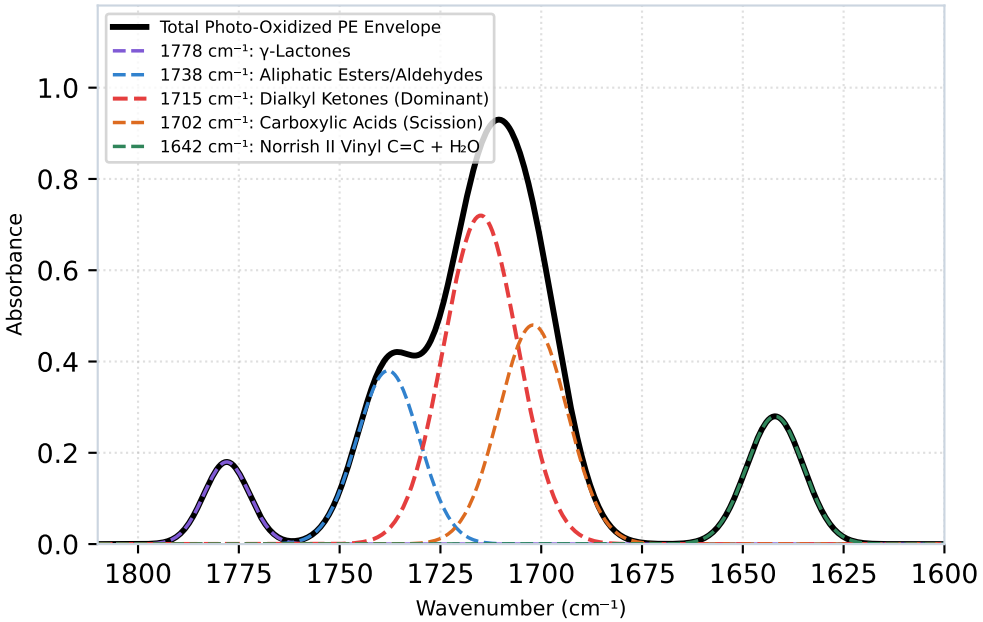
Polymer / Subtype	N–H & C–H Region (3500–2800 cm⁻¹)	Carbonyl / Amide I & II (1760–1500 cm⁻¹)	Upper Fingerprint (1490–1200 cm⁻¹)	Diagnostic Decision Rules (>1200 cm⁻¹)
Nylon 6 (PA6) Polycaprolactam	3300 (s, sharp sym N–H, FWHM<50); 3080 (w-m, Amide B); 2932 & 2858 (s, CH_2)	NO peak at 1735–1700 cm⁻¹! 1638–1642 (vs, sharp Amide I); 1542–1546 (vs, sharp Amide II)	• 1463 & 1438 (m, CH_2 scissor) • 1372 (m, sharp wagging peak) • 1264 (m, sharp Amide III)	PA6 vs PA6,6: PA6 has dominant 1463 scissor + prominent 1372 wag (A_1417/A_1372 < 0.95) and Amide III at 1264 cm⁻¹.
Nylon 6,6 (PA6,6) Polyhexamethylene adipamide	3300 (s, sharp symmetric N–H); 3080 (w-m, Amide B); 2932 & 2858 (s, CH_2 doublet)	NO peak at 1735–1700 cm⁻¹! 1638–1642 (vs, sharp Amide I); 1540–1545 (vs, sharp Amide II)	• 1474 & 1417 (m, sharp crystalline doublet; 1417 adj. to C=O) • 1278 (m, Amide III)	PA6,6 has strong 1474 & 1417 cm⁻¹ doublet with weaker 1372 (A_1417/A_1372 > 1.25) and Amide III shifted to 1278 cm⁻¹.
Polyurethane (PU) (Polyether/Ester Aromatic/Aliph.)	3320–3345 (m-s, broader N–H, blue-shifted +25 cm⁻¹ vs PA); 2955–2855 (m-s, CH_2)	1732 (free) & 1705 (H-bonded) vs Urethane C=O Doublet! 1598 (aro ring), 1532 (vs, Amide II)	• 1412–1415 (m, MDI isocyanate ring) • 1310 (m, C–N) • 1222–1230 (vs, urethane C–O–C)	PU vs Nylon: R_(PU/PA) = A_1715 / A_1640 > 1.5 (Nylon lacks the 1732/1705 urethane C=O doublet!) + strong 1532 & 1225 cm⁻¹ bands.
PAN / Acrylic Fiber (Polyacrylonitrile Copolymer)	2940 & 2870 (m, backbone CH_2) + 2242–2245 cm⁻¹ (vs, sharp nitrile C≡N stretch!)	1732 (m-s, methyl acrylate or vinyl acetate comonomer C=O); ZERO 1600/1500 aromatic bands	• 1454 (s, sharp CH_2 scissor) • 1358 (m, CH wag) • 1250–1230 (m, comonomer C–O)	PAN vs ABS/SAN: Strong 2243 cm⁻¹ C≡N spike + 1732 comonomer C=O with COMPLETE ABSENCE of 3060/3026 and 1601/1493 styrene bands.
Rigid uPVC (Unplasticized PVC)	2972 (m, high-ν CH–Cl stretch!); 2912 (m-s, ν_as CH_2); 2850 (w-m, ν_s CH_2)	Flat 1800–1500 cm⁻¹ (unless dehydrochlorinated polyene around 1630 cm⁻¹)	• 1426–1434 (s, Cl-shifted δCH_2!) • 1332 (m-s, δCH–Cl deformation) • 1254 (s, sharp δCH–Cl in-plane)	Crucial >1200 Rule: Cl inductive effect shifts CH_2 scissor from 1465 down to 1427 cm⁻¹! (A_1427/A_1465 > 2.0) + 1332 & 1254 triad.
Plasticized pPVC (Phthalate/Adipate Plasticized PVC)	2960, 2928, 2872, 2858 (s, ethylhexyl plasticizer CH_3/CH_2 + 2972/2912 PVC)	1724 (vs, phthalate C=O) or 1734 (adipate C=O) + 1600 & 1580 ortho-ring doublet	• 1462 & 1380 (plasticizer alkyl) • 1427 & 1332 (PVC backbone spikes!) • 1278 (vs, broad ester) + 1254 sh.	Separates pPVC from PET/PMMA via ortho-ring 1600/1580 (vs PET 1505/1409) AND the sharp underlying PVC 1427 & 1332 cm⁻¹ peaks.
PDMS / Silicone (Polydimethyl-siloxane)	2963 (s, sharp ν_as Si–CH_3); 2905 (w-m, sharp ν_s Si–CH_3) [NO 2916/2848 CH_2 doublet!]	Completely flat 1800–1450 cm⁻¹ (zero carbonyl, amide, or aromatic bands)	• 1412 (w-m, δ_as Si–CH_3) • 1260 (vs, razor-sharp δ_s Si–CH_3!) • <1220 steep Si–O–Si wing rise	Needle-sharp spike at 1260±2 cm⁻¹ (FWHM < 8 cm⁻¹, A_1260/A_1412 > 3.5) + 2963/2905 C–H doublet.

Interferent Class	3750–2750 cm⁻¹ (OH / NH / CH)	2550–1500 cm⁻¹ (Overtones / Amide / COO⁻)	1500–1200 cm⁻¹ (Bends & Cutoff Edge)	Why It Mimics Plastic & Definitive Rejection Rule
1. Cellulose / Cotton / Linen / Rayon (Natural Fibers)	• 3330–3400 (vs, broad H-bonded OH) • 2895–2902 (m, SINGLE broadish pyranose CH peak; NO 2916/2848!)	• 1635–1645 (w-m, broad adsorbed H_2O bending; no sharp Amide II) • Zero ester C=O (unless acetate)	• 1428 (m, CH_2 sym scissor at C6) • 1370, 1335, 1315 (closely spaced quartet) + steep <1215 C–O edge	Mimics weathered MP or PET. Reject via: (1) Single C–H at 2898 (no 2916/2848 doublet), (2) A_3340/A_2900 > 1.5, (3) 1428/1370/1335/1315 quartet, (4) Bright Blue DANS stain.
2. Proteins / Keratin / Digestive Tissue / Biofilm Matrix	• 3280–3300 (s, broad asymmetric Amide A + OH, FWHM > 100 cm⁻¹) • 2960, 2925, 2872 (WEAK C–H!)	• 1740 (w, lipid ester shoulder) • 1650–1655 (vs, broad Amide I) • 1540–1548 (s, broad Amide II)	• 1450 (m, $\delta CH_2/CH_3$) • 1395–1405 (m, sym COO^- stretch) • 1240 (w-m, broad Amide III)	#1 False Positive for Nylon! Reject via: (1) Low C–H to Amide I (A_2925/A_1650 < 0.35 vs PA > 0.50), (2) High CH_3/CH_2 (A_2960/A_2925 > 0.65), (3) 1400 COO^-, (4) Annulus ΔS ≈ 0.
3. Chitin / Chitosan (Crustacean & Insect Exoskeletons)	• Resolved Triplet: 3445 (s, OH) + 3265 (s, NH) + 3105 (m, Amide B) • 2960, 2930, 2878 (w-m, weak CH)	• 1655 & 1620 (s, SPLIT Amide I doublet in α-chitin!) • 1555 (s, sharp Amide II)	• 1418 & 1378 (m-s, sharp acetamide CH_3 umbrella deformation) • 1310 (m, Amide III) + <1210 edge	Mimics Nylon/PU. Reject via: (1) Split Amide I doublet at 1655 & 1620 cm⁻¹, (2) Resolved 3445/3265 OH/NH pair, (3) Strong 1378 CH_3 with weak 2930 CH_2.
4. Metallic Stearates (Ca / Zn Stearate, Fatty Acid Salts)	• ZERO OH/NH at 3300 cm⁻¹! • 2916 & 2848 (vs, razor-sharp CH_2 doublet identical to PE!)	• NO Amide I at 1640 cm⁻¹! • 1577 & 1540 (s, sharp Ca COO^- doublet) or 1540 (s, Zn COO^-)	• 1468 (m-s, sharp CH_2 scissor) • 1420 (m, sym COO^- stretch) • Progression of weak CH_2 wags	#1 False Positive for PE or Nylon! Reject via: Strong 1577/1540 COO^- band with ZERO 1640 Amide I & ZERO 3300 N–H (rules out PA), while A_1540..1577/A_1468 > 0.5 rules out PE!
5. Carbonates (Calcite / Aragonite Shell Fragments)	• Flat 3000–2800 cm⁻¹ (zero C–H, unless trace organic coating)	• 2512–2515 (w-m, sharp $\nu_1+\nu_3$ spike) • 1795–1798 (m, sharp $\nu_1+\nu_4$ spike!) • Flat 1700–1550 cm⁻¹	• 1420–1455 (vs, MASSIVE broad asymmetric CO_3^{2-} stretch, FWHM > 80 cm⁻¹)	Mimics uPVC (1427) or PC (1775). Reject via: Zero C–H stretches + massive broad 1425 cm⁻¹ carbonate band + unmistakable sharp combination spikes at 1795 & 2512 cm⁻¹.
6. Silicates / Quartz / Clays (Kaolinite, Montmorillonite Silt)	• 3695, 3650, 3620 (sharp structural Al–OH spikes in clays!) • 3400 (broad interlayer H_2O); no CH	• 1995 & 1870 (w, two-humped Si–O combination bands in quartz) • 1630 (w-m, H–O–H bend)	• Flat 1550–1300 cm⁻¹ • <1260 cm⁻¹ STEEP MONOTONIC CLIFF into 1200 cutoff (Si–O wing)	Dominant sediment interference. Reject via: Sharp >3600 clay spikes or 1995/1870 quartz humps + zero C–H + Cutoff Ramp Ratio A_1210 / A_1300 > 2.5.
7. 'Green Dot' Lab Contaminant Artifact	• Completely flat 4000–1800 cm⁻¹! • Zero O–H, N–H, or C–H stretches (SNR_CH < 1.8)	• Onset of smooth arch around 1800–1750 cm⁻¹ (no resolved carbonyl or amide peaks)	• Single broad, structureless bell-shaped hump spanning 1800–1200 cm⁻¹ (FWHM > 220 cm⁻¹)	Looks like green PE/PP dot (15–35 µm) in DANS fluo! Reject via: SNR_CH < 2.0 AND 2nd derivative roughness $\|\|d^2A/d\nu^2\|\|$_(1750–1250) ≈ 0 (zero sharp vibrational peaks).

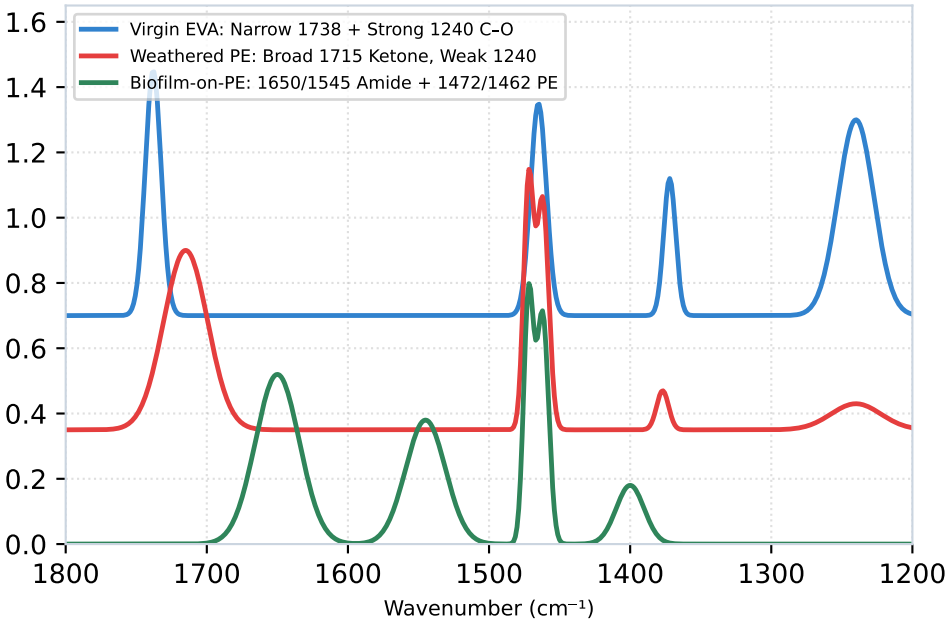
Critical 4-Way Disentanglement Matrix: Resolving the '1640 / 1540 cm⁻¹ Amide / Carboxylate Confusion Zone'

Diagnostic Metric	Synthetic Nylon (PA6 / PA6,6)	Biological Protein / Tissue / Keratin	Chitin (Crustacean / Insect)	Ca / Zn Stearate (Lubricant / Salt)
N–H / O–H Profile (3500–3050 cm⁻¹)	Single SHARP symmetric peak at 3300 cm⁻¹ (FWHM < 55) + 3080	Broad asymmetric band at 3285 cm⁻¹ (FWHM > 100 cm⁻¹)	Resolved Triplet: 3445 (OH), 3265 (NH), 3105 cm⁻¹	COMPLETELY ABSENT (Flat baseline at 3300 cm⁻¹)
Aliphatic C–H to Amide/COO^- Ratio	HIGH: A_2932 / A_1640 ∈ [0.50, 0.95] (Strong 2932 & 2858 CH_2 doublet)	LOW: A_2925 / A_1652 ∈ [0.10, 0.35] (Weak C–H vs massive Amide I)	LOW: A_2930 / A_1655 ∈ [0.15, 0.35] (Weak C–H stretches)	VERY HIGH: A_2916 / A_1540 > 1.2 (Razor-sharp PE-like 2916/2848)
CH_3 to CH_2 Stretch (A_2960 / A_2928)	VERY LOW (< 0.25) Purely polymethylene -(CH_2)_n-	HIGH (0.65 – 1.10) Rich in Leu/Val/Ala CH_3 side chains	MODERATE (0.50 – 0.85) Acetamido CH_3 + pyranose CH	VERY LOW (< 0.20) Linear C_{18} stearoyl chain
1660–1530 cm⁻¹ Doublet Geometry	Sharp 1640 (Amide I) + Sharp 1544 (Amide II)	Broad 1652 (Amide I) + Broad 1545 (Amide II)	SPLIT Amide I (1655 & 1620) + Sharp 1555 (Amide II)	ZERO 1640 peak! Only 1577/1540 (Ca) or 1540 (Zn) COO^- spike!
1480–1360 cm⁻¹ Confirming Marker	1463/1372 (PA6) or 1474/1417 (PA6,6) CH_2 scissor	1450 bend + 1400 cm⁻¹ symmetric COO^- stretch	Strong sharp 1378 cm⁻¹ acetamido CH_3 umbrella	1468 sharp CH_2 scissor + 1420 cm⁻¹ sym COO^- stretch

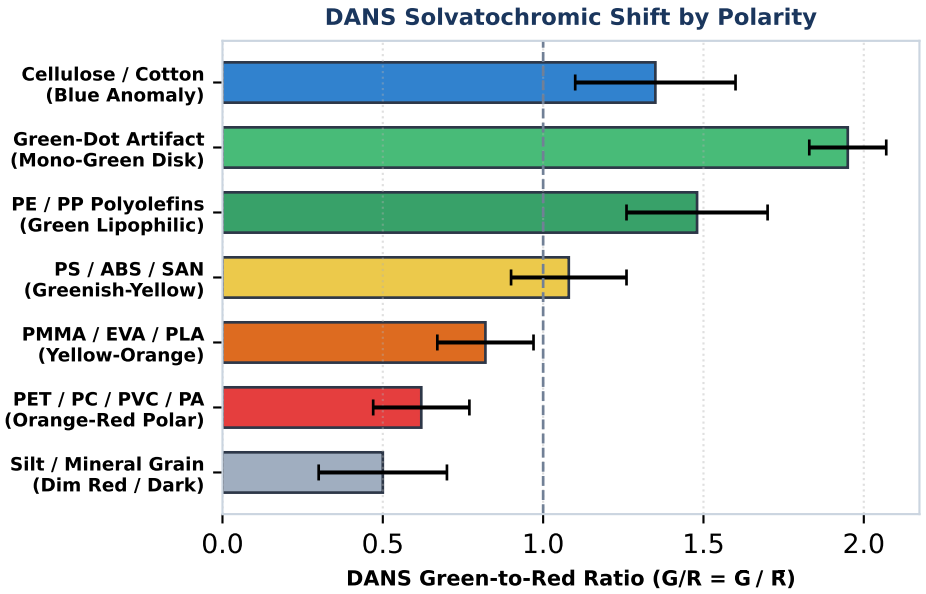
Deconvolution of the Photo-Oxidative Carbonyl & Vinyl Complex

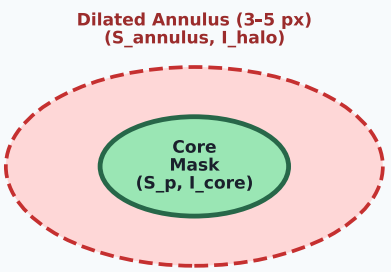


Disentangling Virgin EVA vs. Abiotic Oxidized PE vs. Biotic Biofilm-PE



Degradation / Confusion Case	Spectral Band Formula / Ratio (4000–1200 cm⁻¹)	Severity / Decision Thresholds	Physical & Chemometric Rationale
1. Polyolefin SAUB Carbonyl Index (CI) [PE & PP]	CI_PE = Area(1800–1680) / Area(1500–1420) CI_PP = Area(1800–1680) / Area(1485–1420) (Normalized to δCH₂/CH₃, NOT saturating 2916!)	• Virgin: CI < 0.05 • Mild Weathering: 0.05 ≤ CI < 0.30 • Moderate: 0.30 ≤ CI < 0.80 • Severe Oxidation: CI ≥ 0.80	In FPA transmission, 2916/2955 cm⁻¹ C–H stretches saturate in particles >15 µm thick. Normalizing to 1463/1456 cm⁻¹ provides a thickness-linear oxidation metric.
2. Hydroxyl & Vinyl Weathering Indices (HI & VI)	HI = Area(3600–3150) / Area(1500–1420) VI = A_1642 / A_1463	• Virgin PE/PP: HI < 0.08, VI < 0.04 • Photo-oxidized: HI > 0.20, VI > 0.08 (Norrish Type II chain scission)	Tracks alcohol/hydroperoxide accumulation (3380 cm⁻¹) and terminal vinyl group formation (1642 cm⁻¹) during UV exposure.
3. Weathered PE vs. Virgin EVA Copolymer	Compare ν_max(C=O), FWHM(C=O), Acetate Ratio R_ac = A_1240 / A_C=O, and Hydroxyl Index HI	• Virgin EVA: ν_max = 1738±2 cm⁻¹, FWHM < 22, R_ac = A_1240/A_1738 ∈ [0.48, 0.95], HI < 0.08 • Weathered PE: ν_max = 1712–1718 cm⁻¹, FWHM > 30, R_ac = A_1240/A_1715 < 0.22, HI > 0.15	In EVA, every 1738 C=O group has an attached 1240 C–O–C bond. In oxidized PE, ketones (1715) and acids (1702) dominate without proportional 1240 cm⁻¹ ester C–O.
4. Biofilm-Coated PE/PP vs. True Nylon (PA) vs. Polyurethane (PU)	Inspect C–H Stretch Center (2916 vs 2932), 1472/1462 PE Scissor, 3300 N–H Width, and Local Annulus Differential ΔS	• Biofilm-on-PE: C–H locked at 2916/2848 + 1472/1462 scissor + broad 3350 OH/NH + 1650/1545 biofilm • True Nylon: C–H shifted to 2932/2858 + sharp 3300 N–H • Local Annulus: ΔS = S_p - S_annulus strips matrix protein!	Prevents environmental PE/PP coated in proteinaceous biofilm or digestive tissue from being misclassified as Nylon or PU.
5. Weathered PS / ABS vs. PET / Acrylate Copolymers	Compare Ring Spike Position (1493 vs 1505), Carbonyl Center (1720–1690), and 2237 cm⁻¹ Nitrile Spike	• Oxidized PS/ABS: Retains sharp 1493 & 1452 cm⁻¹ monosubstituted ring doublet + broad 1715–1690 C=O (plus 2237 C≡N in ABS) • PET: Ring spike is at 1505 & 1409 cm⁻¹ (+12 cm⁻¹)	In ABS, the polybutadiene phase oxidizes rapidly in sunlight, creating a 1715 cm⁻¹ carbonyl band while the SAN matrix retains 2237 and 1493/1452 cm⁻¹.





1. Optical Edge Sharpness:
   C_edge = (I_core - I_halo) / (I_halo + ε)
   • Real Particle: C_edge ≥ 0.35
   • Diffuse YOLO Glow: C_edge < 0.22 → Reject!

2. FPA Chemical Map Alignment:
   ρ_spatial = Corr(Mask_fluo, Map_FPA)
   • Confirms IR absorber matches Fluo shape

3. Local Annulus Spectral Subtraction:
   ΔS(ν) = S_core(ν) - S_annulus(ν)
   • Eliminates adjacent massive absorber bleed & background protein mimicry!

Concept ID & Name	Mathematical Definition	Expert Perceptual Meaning	Active Threshold / Gate	Cross-Modal Interaction Rule in CBM / LLM
O1: `fluo_particle_sharpness` (Visible Particle vs. Diffuse Halo)	C_edge = (I_core - I_halo) / (I_halo + 5.0) + Sobel boundary gradient along mask contour	Expert Check #1: Is there a crisp physical particle in the crop, or just a diffuse fluorescence cloud?	Visible Particle: C_edge ≥ 0.35 Diffuse Glow: C_edge < 0.22 (Tag: `Diffuse Stain`)	If C_edge < 0.22 AND spectral SNR < 3.5: Immediately boost `Non-Plastic / Diffuse Halo` and suppress top-1 polymer confidence.
O2: `dans_lipophilic_green` (Non-Polar Polyolefin Stain Signature)	G/R = G_core / (R_core + ε); SSI = (G - R)/(G + R) with Blue Fraction < 0.28	Expert Check #2a: Greenish to green-yellow fluorescence typical of non-polar PE, PP, and PS.	Greenish: G/R ∈ [1.15, 2.20] and I_core > 25 counts (Tag: `Green Fluor`)	Corroborates PE/PP/EPR when 2916/2848 or 2955/1376 bands are present. If PE spectrum has G/R < 0.65 (deep red), flag as `Color Discrepancy`.
O3: `dans_polar_orange_red` (Polar Ester / Amide / Chlorinated Signature)	G/R = G_core / (R_core + ε) with R_core > G_core and I_core > 20 counts	Expert Check #2b: Yellow-orange to red solvatochromic shift in polar PET, PA, PU, PC, PMMA, PVC.	Orange/Red: G/R ∈ [0.35, 0.95] (Tags: `Yellow Fluor`, `Orange Fluor`, `Red Fluor`)	Corroborates carbonyl/amide polymers. If weak dim red (I_core < 18) with cutoff ramp <1250 cm⁻¹, boost `Mineral Silt / Non-MP`.
O4: `blue_cellulose_anomaly` (Blue Fiber Stain vs. Synthetic PET)	Blue_Frac = B_core / (R + G + B) coupled with Fiber Aspect Ratio AR = L_maj / L_min	Expert Check #2c: Strong blue fluorescence on a fiber indicates cellulose/cotton, rarely PET!	Blue Anomaly: Blue_Frac > 0.36 (or Cyan/Blue Hue) + AR > 2.5 (Tag: `Blue Fluor`, `Fiber`)	Contradiction Gate: Even if spectrum has ester-like bands, penalize PET logit and elevate `Cellulose / Natural Fiber` into the conformal proposal set!
O5: `green_dot_morphology` ('Green Dot' Lab Contaminant Cues)	Circularity = 4π·Area / Perim² > 0.82, ESD ∈ [12, 38] µm, Mono-Green G/R > 1.65, Solidity > 0.92	Expert Check #3: Distinct round green dot that mimics PE/PP optically but has 1800–1200 hump.	Morph + Color Match > 0.80 coupled with Spectral Concept S20 (`hump_1800_1200`)	When O5 (`green_dot_morphology`) AND S20 (broad 1800–1200 hump + zero C–H stretch) fire together, route 100% to `Contaminant (Green Dot)`.
O6: `fpa_spatial_corr` (Chemical Image to Fluo Mask Match)	ρ_spatial = Max_band Corr(M_fluo, A_FPA(band)) across CH (2920), C=O (1720), Amide (1640)	Expert Check #4a: Does a matching particle appear in the FPA chemical heatmaps at the mask position?	Strong Match: ρ_spatial ≥ 0.45 Unverified Position: ρ < 0.20 (widens conformal set)	Weight pixel spectra by FPA chemical intensity; if ρ_spatial < 0.20, lower effective visibility v_k so conformal set reflects registration uncertainty.
O7: `neighbor_absorber_bleed` (Adjacent Strong Absorber Artifact)	R_bleed = Max(A_annulus) / (Max(A_core) + ε) in 1800–1200 cm⁻¹	Expert Check #4b: Is background subtraction corrupted by a nearby highly absorbing object/clump?	Clean Surroundings: R_bleed < 0.5 Severe Bleed: R_bleed > 1.2 (Tag: `Under/Over-Subtracted`)	Triggers Local Annulus Differential spectrum ΔS and alerts analyst that baseline distortion is caused by an adjacent big absorber.
O8: `annulus_amide_silt_match` (Background Mimicry Verification)	CosSim(S_core[1700:1200], S_annulus[1700:1200])	Expert Check #5: Are putative 'Nylon' or 'Silt' bands identically present in neighboring dark pixels?	Background Mimic: Sim > 0.85 and ||ΔS||_2 < 0.15 (Tag: `Likely Bkgd`)	Solves the Nylon verification bottleneck: if Amide bands match the surrounding non-fluorescent filter, penalize Nylon and route to `Background Matrix`.

ID	Feature Tag / Concept Name	Peak Window (cm^{-1})	Baseline Anchors	Target Polymers / Classes	Perceptual Breakpoint & Discerning Criterion
S01	`PE_CH2_Doublet` *(Existing App #0)*	2916±6 & 2848±5 (Trough: 2880)	3010 & 2780 cm^{-1}	PE (HDPE, LDPE, LLDPE), EVA, Stearates	Resolved bimodal split ($d^2A/d\nu^2$ minima at 2916 & 2848) with A_2955/A_2916 < 0.20. Saturation flag if A_2916 > 1.4 AU.
S02	`PP_CH_Quartet` *(Existing App #1)*	2955, 2918, 2870, 2838	3010 & 2780 cm^{-1}	PP, EPR/EPDM, PP/PE Copolymer	4-peak multiplet with A_2955/A_2918 > 1.05 (PP) or prominent 2955/2870 shoulders (EPR) and $\nu_s(CH_2)$ red-shifted to 2838 cm^{-1}.
S03	`PE_Scissor_Split` *(New Subtype Gate)*	1472±3 & 1462±3	1500 & 1440 cm^{-1}	HDPE vs. LDPE/LLDPE Subtype Separation	Orthorhombic crystalline split: A_1472/A_1462 ≥ 0.90 indicates HDPE; single/asymmetric 1463 peak indicates LDPE/EVA.
S04	`CH3_Umbrella_1377` *(Existing App #2)*	1376–1378 (vs 1368/1352 wag)	1405 & 1335 cm^{-1}	PP vs. PE, and LDPE vs. HDPE	R_bend = A_1377/A_1460: >0.85 (PP), 0.30–0.75 (EPR), 0.12–0.25 (LDPE branch spike > 1368 wag), <0.05 (HDPE, flat at 1377).
S05	`Aromatic_CH_Fingers` *(New Styrenic Gate)*	3082, 3060, 3026	3120 & 2995 cm^{-1}	PS, ABS, SAN, PET, PC	Sharp multiplet >3000 cm^{-1}. Ratio A_2924/A_3026 separates SAN (1.0–1.5) from ABS (1.7–3.2, rubber CH_2 boost).
S06	`Styrene_Ring_1493_1452` *(Existing App #5)*	1601, 1493±3, 1452±3	1625 & 1420 cm^{-1}	PS, ABS, SAN (vs PET/PC 1505!)	Sharp spikes at 1493 & 1452 cm^{-1} (FWHM < 12 cm^{-1}). Replaces lost 756/698 cm^{-1} bands! Separates PS (1493) from PET/PC (1505).
S07	`Nitrile_Spike_2237` *(Existing App #6)*	2237–2244	2275 & 2200 cm^{-1}	ABS, SAN (2237), PAN Fiber (2243)	Isolated sharp spike in silent window. Paired with 1493 ring → ABS/SAN; paired with 1732 C=O & zero ring → PAN acrylic fiber.
S08	`Carbonyl_Position_Shift` *(Expanded App #3)*	1775, 1752, 1738, 1730, 1715	1815 & 1675 cm^{-1}	PC (1775), PLA (1752), EVA (1738), PMMA (1730), PET/PBT (1715)	Continuous peak-center & FWHM tracker: resolves carbonate (1775), α-ester (1752), acetate (1738), acrylate (1730), terephthalate (1715), and broad weathered ketone/acid (1715/1702).
S09	`Terephthalate_1505_1409` *(New Polyester Gate)*	1578, 1505±3, 1409±3, 1342	1610 & 1315 cm^{-1}	PET vs. PBT vs. PC	Para-substituted ring triad (1578/1505/1409). Strong 1342 trans wag identifies PET; strong 1460/1386 identifies PBT.
S10	`PC_Spike_1506_GemCH3` *(New PC Gate)*	1506±2 (vs), 1386 & 1364	1540 & 1340 cm^{-1}	Polycarbonate (PC)	Needle-sharp 1506 cm^{-1} ring spike (A_1506/A_1775 > 0.50) + 1386/1364 cm^{-1} Bisphenol-A gem-dimethyl doublet + 1230 C–O.
S11	`PMMA_OCH3_Doublet` *(New Acrylic Gate)*	2995/2950/2842 & 1449/1435, 1388	1510 & 1365 cm^{-1}	PMMA vs. PLA	PMMA: 2842 sym O–CH_3 + 1449/1435 bend + 1730 C=O. PLA: 2997/2946 CH_3 + 1453 & 1383/1360 doublet + 1752 C=O.
S12	`EVA_Acetate_1240_1372` *(New EVA vs Weath PE)*	1738, 1372, 1240±8	1800 & 1205 cm^{-1}	Virgin EVA vs. Weathered PE	Proportional acetate C–O–C band at 1240 cm^{-1} (A_1240/A_1738 > 0.45) + sharp 1372 CH_3 bend confirms EVA over oxidized PE.
S13	`Urethane_Split_CO` *(New PU vs PA Gate)*	1732 & 1705 C=O + 1532 + 1225	1770 & 1205 cm^{-1}	Polyurethane (PU) vs. Nylon (PA)	H-bonded/free urethane C=O doublet (1732/1705) with A_1715/A_1640 > 1.5 separates PU from Nylon (which has zero 1715).
S14	`Amide_I_II_Geometry` *(Expanded App #4)*	3300 NH, 1640–1655 Amide I, 1545 II	1700 & 1490 cm^{-1}	Nylon (PA6/PA6,6) vs. Protein vs. Chitin	Nylon: sharp 3300 NH + 2932/2858 CH_2 (A_2932/A_1640 > 0.50). Protein: broad 3285 + weak CH + 1400 COO^-. Chitin: 1655/1620 split.
S15	`PA6_vs_PA66_Cryst` *(New Polyamide Subtype)*	1474/1417/1278 vs. 1463/1372/1264	1495 & 1240 cm^{-1}	Nylon 6,6 vs. Nylon 6	PA6,6: strong 1474 & 1417 doublet (A_1417/A_1372 > 1.25) + 1278. PA6: dominant 1463 scissor + sharp 1372 wag + 1264 Amide III.
S16	`PVC_Chlorine_Triad` *(New >1200 PVC Gate)*	2972, 1427±3, 1332, 1254±3	1480 & 1220 cm^{-1}	Rigid uPVC & Plasticized pPVC	Replaces <700 cm^{-1} C–Cl! Cl shifts CH_2 scissor to 1427 cm^{-1} (A_1427/A_1465 > 2.0) + 1332 & 1254 CH–Cl bends + 2972 stretch.
S17	`PDMS_Spike_1260` *(New Silicone Gate)*	2963/2905, 1412, 1260±2 (needle)	1300 & 1230 cm^{-1}	PDMS / Silicone Elastomer	Razor-sharp symmetric Si–CH_3 umbrella spike at 1260 cm^{-1} (FWHM < 8 cm^{-1}, A_1260/A_1412 > 3.5) + 2963/2905 doublet.
S18	`Stearate_Carboxylate` *(New Contaminant Gate)*	2916/2848 PE CH_2 + 1577 & 1540 COO^-	1620 & 1500 cm^{-1}	Ca / Zn Stearate Fatty Acid Salts	Prevents false PE/Nylon calls! Sharp 2916/2848 CH_2 + 1577/1540 carboxylate with ZERO 1640 Amide I and ZERO 3300 N–H.
S19	`Cellulose_Pyranose_Quartet` *(New Natural Fiber Gate)*	3340 OH, single 2898 CH, 1428/1370/1335/1315	3000 & 1280 cm^{-1}	Cellulose / Cotton / Rayon Fibers	Single broadish C–H at 2898 (no 2916/2848 doublet!) + broad 3340 OH + 1428/1370/1335/1315 progression + <1215 C–O wing.
S20	`Mineral_or_GreenDot_Ramp` *(New Artifact Gate)*	3695/3620, 2512/1795, 1800–1200 hump/ramp	3750 & 1200 cm^{-1}	Clays, Quartz, Calcite, 'Green Dot' Artifact	Calcite: 1425 broad + 1795/2512 spikes. Clay/Silt: 3695/3620 + <1250 cliff. Green Dot: zero C–H + smooth 1800–1200 bell hump.

TIER 1 -> TIER 3: CHEMOMETRIC REPORT CARD (JSON)

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    "S08_Carbonyl":   {"wn": 1718, "snr": 3.1, "v_k": 0.62},
    "S09_PET_Ring":   {"snr": 1.1, "v_k": 0.09, "st": "NOISE"},
    "S19_Cellulose":  {"snr": 4.8, "v_k": 0.89, "st": "VISIBLE"},
    "OH_3340_Broad":  {"snr": 7.4, "v_k": 0.98, "st": "VISIBLE"},
    "Single_CH_2898": {"snr": 4.2, "v_k": 0.84, "st": "VISIBLE"}
  },
  "optical_and_spatial_concepts": {
    "O1_Sharpness":   {"c_edge": 0.78, "st": "SHARP_PARTICLE"},
    "O4_Blue_Fiber":  {"blue_frac": 0.44, "ar": 4.6, "flag": true},
    "O6_FPA_Spatial": {"rho": 0.67, "bleed_ratio": 0.18},
    "O8_Annulus_Diff": 0.82
  }
}
```

TIER 3 & 4: LOCAL LLM OUTPUT + CONFORMAL SET

```
{
  "auto_tags": [
    "Blue Fluor", "Fiber", "High SNR",
    "Cellulose: Single C-H (2898) + Broad OH (3340)",
    "Contradiction: PET Cosine vs Blue Fiber"
  ],
  "expert_chain_of_thought": "Although 24-D cosine similarity
   ranks PET #1 (0.79) due to a weak 1718 cm⁻¹ shoulder (likely
   dye/finish ester), the terephthalate ring triad (1505/1409)
   is absent (SNR 1.1 < θ_k), C-H shows a single 2898 cm⁻¹ peak
   with broad 3340 cm⁻¹ OH and 1428/1370/1335/1315 quartet, and
   DANS fluorescence is bright blue (Blue_Frac=0.44) on a fiber
   (AR=4.6). Classic dyed cellulose/cotton fiber, not PET.",
  "calibrated_posteriors": {
    "Non-Plastic / Cellulose Fiber": 0.89,
    "PET": 0.07,
    "PU": 0.04
  },
  "conformal_set_95": ["Non-Plastic / Cellulose Fiber"],
  "conformal_status": "SINGLETON_REJECT_NON_MP"
}
```

PROJECT DELIVERABLES & HOW TO OPEN THIS MANUAL IN YOUR REMOTE ANTIGRAVITY SESSION

1. Primary Project Folder (Data Drive):
   - PDF Manual: D:\user_data\smolen\261012_Perceptual_Chemometric_Manual_and_Roadmap\Perceptual_Chemometric_Manual_and_Roadmap.pdf
   - Agent Markdown Ref: D:\user_data\smolen\261012_Perceptual_Chemometric_Manual_and_Roadmap\AGENT_CHEMOMETRIC_REFERENCE.md
   - JSON Feature Spec: D:\user_data\smolen\261012_Perceptual_Chemometric_Manual_and_Roadmap\perceptual_feature_catalog_4000_1200.json
   - High-Res Page PNGs: D:\user_data\smolen\261012_Perceptual_Chemometric_Manual_and_Roadmap\rendered_pages\page_01.png .. page_12.png

2. Three Ways to View the PDF Directly Inside Your Remote Antigravity Session:
   - Method A (Direct IDE Tab): Click the [Perceptual_Chemometric_Manual_and_Roadmap.pdf] link in chat or open `docs/Perceptual_Chemometric_Manual_and_Roadmap.pdf` in the Antigravity file explorer (uses built-in PDF viewer).
   - Method B (Interactive Artifact Carousel): Open the Antigravity Artifact `Perceptual_Chemometric_Manual_Viewer.md` in the side panel to flip through all 12 rendered pages inline with zero external viewer dependencies.
   - Method C (Web Annotator Static Route): Served at `http://localhost:8000/static/Perceptual_Chemometric_Manual_and_Roadmap.pdf` whenever the Web Annotator or Perceptibility App server is running.